

AREESHA NAEEM

Junior Software Developer | AI, Computer Vision & Full-Stack Development
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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in software development, AI/ML, computer vision, data processing, and full-stack applications. Strong foundation in Python, data structures, databases, software engineering, and machine learning, with practical experience developing end-to-end AI and software systems. Developed research-oriented prototypes including MKEA-Lite, a multilingual audio knowledge engineering system, and an Assistive Object Detection System using YOLO, OpenCV, Streamlit, and text-to-speech. Experienced in API integration, structured experimentation, technical documentation, and Git/GitHub-based collaborative development.

TECHNICAL SKILLS

Programming & Computer Science: Python, JavaScript, Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Design Principles

AI, ML & Computer Vision: Machine Learning, Computer Vision, YOLO, OpenCV, Scikit-learn, Natural Language Processing, Speech-to-Text, Language Detection, Semantic Embeddings, Named-Entity Extraction

Data & Search: Pandas, NumPy, SQL, Data Cleaning, Exploratory Data Analysis (EDA), FAISS, Semantic Search

Software Development: React.js, Next.js, REST APIs, Django/Flask, Streamlit, HTML5, CSS3, Tailwind CSS, Debugging & Testing

Knowledge Representation: NetworkX, Knowledge Graphs, Structured Metadata, Entity-Relationship Representation

Tools & Collaboration: Git, GitHub, Branching, Pull Requests, Code Reviews, Technical Documentation

PROFESSIONAL EXPERIENCE

Full Stack Developer | DriveThru (Remote)

May 2026 – Present

- Collaborate with a cross-functional development team to build and maintain frontend and backend features using modular software development practices.
- Develop and optimize course and university filtering functionality using modern web technologies and backend services.
- Debug and resolve UI and functional issues to improve application reliability and usability.
- Integrate and work with REST APIs and structured application data across frontend workflows.
- Manage development through Git/GitHub workflows, including feature branches, pull requests, code reviews, and implementation of review feedback.
- Communicate technical requirements and coordinate with team members to deliver assigned development tasks within deadlines.

Computer Science Intern | Pakistan Aeronautical Complex (PAC), Kamra

Jul 2025 – Aug 2025

- Supported software development and technical workflows within an engineering and aerospace environment.
- Worked with frontend interfaces, REST APIs, databases, and software tools supporting internal engineering and operational workflows.
- Cleaned, transformed, and analyzed datasets using Python and SQL to improve data quality and reporting.
- Collaborated with engineering teams on software and technical tasks and contributed to technical reporting and documentation.
- Applied software development, database, data-processing, and problem-solving skills to practical engineering projects.

RESEARCH & AI PROJECTS

MKEA-Lite — Multilingual Audio Archive Knowledge System

Python, Whisper, FAISS, NetworkX, Streamlit

- Developed a research-oriented prototype that converts audio recordings into searchable, structured knowledge through an end-to-end processing pipeline.
- Implemented Whisper-based speech recognition and automatic language detection, producing timestamped transcript segments from uploaded audio.
- Applied multilingual Sentence Transformer embeddings and FAISS-based semantic search for natural-language retrieval.
- Extracted named entities from transcripts and represented relationships between recordings, transcript segments, and entities using a NetworkX knowledge graph.
- Built a Streamlit interface integrating audio upload, transcription, processing, semantic retrieval, and knowledge-graph visualization/summary.

Assistive Object Detection System

Python, YOLO, OpenCV, Streamlit, pyttsx3

- Developed a computer vision prototype using a pretrained YOLO model to detect objects and report confidence scores from uploaded images.
- Implemented approximate left, center, and right spatial positioning to generate structured assistive scene descriptions.
- Built an interactive Streamlit interface with text-to-speech feedback for audio-based scene information.

AI-Based Course Registration System

React, Python, Machine Learning, SQL

- Developed an AI-assisted course registration system that recommends suitable courses based on students' academic information and preferences.
- Implemented structured data processing and machine learning techniques to generate relevant course recommendations.
- Integrated frontend components with backend services and SQL-based data handling through modular software components.

Customer Churn Prediction System

Python, Pandas, NumPy, Scikit-learn

- Built an end-to-end machine learning pipeline for customer churn prediction, including preprocessing, EDA, feature engineering, model training, and evaluation.

Breast Cancer Classification System

Python, Scikit-learn

- Developed a machine learning classification system with data preprocessing, feature scaling, model training, and evaluation using Scikit-learn.

EDUCATION

Bachelor of Science in Computer Science | COMSATS University Islamabad, Attock Campus

CGPA: 3.52 / 4.00

Sep 2022 – Jun 2026

CERTIFICATIONS & ACHIEVEMENTS

- **3rd Position — University Coding Competition:** Recognized for programming, problem-solving, and algorithmic coding skills.
- **Prime Minister's Merit-Based Laptop Award:** Awarded under the merit-based laptop scheme in recognition of academic performance.
- **IBM Data Analysis with Python:** Cognitive Class Certification.